

Delay Shield

Product Requirements Document

1. Problem Statement

When a Zomato delivery runs late, users receive no communication from the platform. The ETA passes. The progress bar freezes. The app offers nothing — no updated time, no reason, no acknowledgment. The user's only options are to refresh the app repeatedly or call support.

The frustration is not about the delay. Delays are operationally inevitable. The frustration is the silence. A user who knows their food is 15 minutes late because of traffic is a patient user. A user who knows nothing becomes an anxious user, then an angry user, then a churned user.

> Data shows delivery delay appears in 48% of all negative reviews — the single largest driver of poor ratings ahead of food quality and customer support. Post-delay order ratings average 2.1 out of 5.0. These are not outliers. This is the median Zomato experience for a delayed order.

What the data shows

Category	Finding	So What
Top pain point	48% of negative reviews mention delay explicitly	No other issue comes close — fixing this has the biggest lever
Support ticket volume	~34% of all support tickets are delay-related	Each ticket has an operational cost. A notification eliminates it.
Post-delay rating	Average 2.1/5.0 among users who mention delay	One bad delay turns a satisfied user into a 1-star reviewer
Negative sentiment overall	41.4% of 5,000 reviews are negative	Platform health is at risk without addressing root causes
Churn risk	9.7% of user base flagged as critical churn risk	These users are one more bad experience away from leaving

Why now

- Swiggy has started sending ETA update notifications. Zomato has nothing comparable. The gap is getting wider.
- Delay-related support tickets are estimated at 34% of total volume. Proactive communication eliminates the ticket before it's raised.
- User tolerance for silence during service failures has dropped. A push notification is now a baseline expectation, not a nice-to-have.

2. User Personas

Three archetypes were identified from review pattern analysis. Each has a different relationship with delivery timing and a different threshold for churn.

	Arjun, 26 — Bangalore	Priya, 34 — Mumbai	Rahul, 20 — Pune
Role	Software engineer — orders lunch daily during his 30-minute office break	Working mother — orders dinner for family 3–4x per week, Rs 600–900 per order	Engineering student — orders late night 10 PM to 2 AM, highly price-sensitive
Core need	Food arrives within 30 mins. Every minute over that costs him his break.	Reliable dinner before kids' bedtime. Timing is non-negotiable.	Food before restaurants close. No fallback option exists at 1 AM.
Pain with delay	App gives him nothing. He opens it 5–6 times, feels ignored, eats at his desk.	Hungry kids, ruined routine, high stress. Most likely to call support and leave a review.	Restaurants close during the wait. He has no backup and vents on Twitter.
Churn trigger	Two bad deliveries in one week	One cold meal with zero explanation	Two consecutive late-night failures
Priority	**High**	**Critical**	**High**

3. Feature Specification

What Delay Shield does

Delay Shield detects when a delivery has breached its promised ETA and fires a proactive response — a push notification explaining what's happening, an in-app modal with compensation — before the user opens support, before they refresh the app for the fifth time, before they lose trust.

The goal is not to apologise for a delay. It's to get ahead of the user's anxiety before it becomes frustration. A user who hears from Zomato first is a user who feels in control.

The prototype

Designed in Figma. The Before screen shows the current experience: a delay banner with no response from the platform. The After screen shows Delay Shield active: a modal with a simple message, a cash credit, and a clear next action.

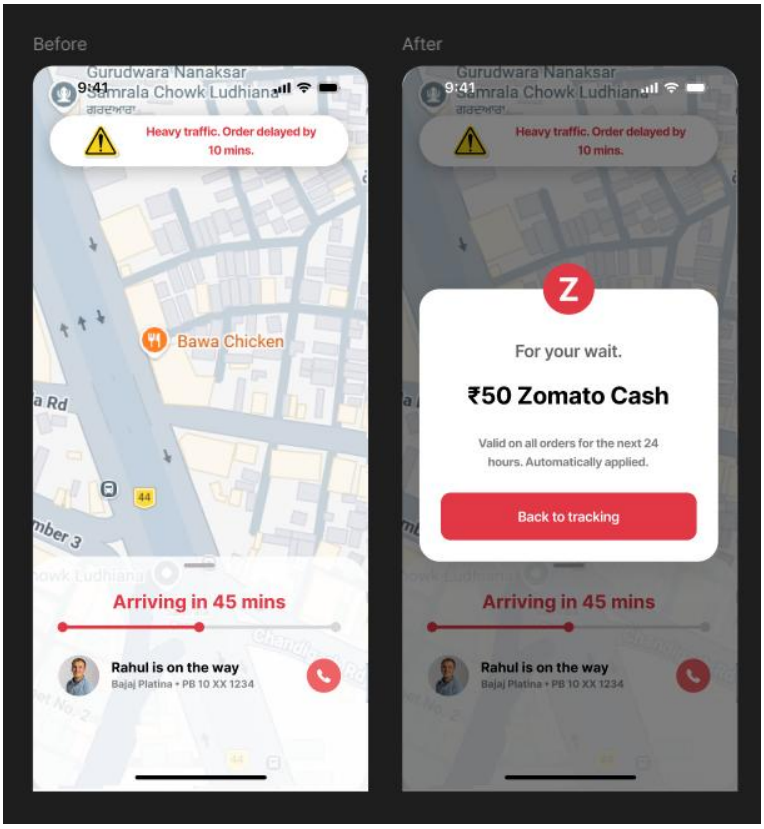


Figure 1. Delay Shield — Before (left) and After (right). Figma prototype.

Before (current state)

- Delay banner fires: "Heavy traffic. Order delayed by 10 mins."
- ETA shown in red: "Arriving in 45 mins"
- Nothing else happens. No explanation, no compensation, no follow-up. The user is on their own.

After (Delay Shield active)

- "For your wait." — three words. Not a corporate apology. Just a human acknowledgment.
- "Rs 50 Zomato Cash" — auto-credited, no code required, valid for 24 hours on any order.
- "Back to tracking" CTA returns the user to the live map. One tap, no friction.

Trigger conditions

Delay Shield fires only when all seven conditions are met simultaneously. Any single condition failing means no trigger.

Condition	Rule	Why
Order status	IN_TRANSIT and accepted by delivery partner	Excludes pre-pickup restaurant delays — separate problem
ETA breach	Current time > original ETA by more than 5 minutes	Buffer prevents false positives on near-on-time orders
Not yet delivered	Status is not DELIVERED or CANCELLED	Notification after delivery is pointless
No active support ticket	No open ticket for this order in last 20 minutes	Prevents double compensation for users who already escalated
No prior compensation	Compensation Issued = false for this order Id	One compensation per order. No exceptions.
User reachable	Push notifications enabled OR app is currently open	Fallback to in-app banner if push is disabled
Min order value	Subtotal >= Rs 100	Prevents abuse via low-value orders

Step-by-step flow

1. Delivery partner accepts order. Backend starts tracking against the promised ETA.
2. At T+5 minutes past ETA, backend checks all seven trigger conditions.

3. All pass: system fetches delay reason from delivery partner app via Reason API.
4. Reason is mapped to a plain-language message per the table in Section 3.4.
5. Push notification fires: "[Restaurant] order is running a bit late. [Reason]. New ETA: [time]"
6. User opens app or app is open. Delay Shield modal appears automatically over the map.
7. User taps "Back to tracking". Modal dismisses. Rs 50 Zomato Cash credited in background.
8. compensationIssued flag set to true for this orderId. No second trigger possible.
9. Post-delivery: standard rating prompt appears. Post-delay rating is the North Star KPI.

Delay reason mapping

If the Reason API fails for any reason, the fallback copy fires. The user never sees a blank field.

System Reason	What the user sees
TRAFFIC_HIGH	"Heavy traffic on the route — your partner is on their way."
RESTAURANT_DELAYED	"The kitchen is taking a little extra time with your order."
WEATHER_RAIN	"Rain is slowing things down — your partner is being careful."
PARTNER_SEARCHING	"We're finding you the closest delivery partner — almost there."
UNKNOWN / API failure	"There's an unexpected delay — we're on it and keeping you posted."

Compensation tiers

Delay duration	Trigger	Compensation
5 to 15 minutes	Push notification only	No credit. Just a heads-up.
15 to 30 minutes	Notification + modal	Rs 30 Zomato Cash — auto-credited, valid 24h
30 to 60 minutes	Full Delay Shield	Rs 50 Zomato Cash — auto-credited, valid 24h
More than 60 minutes	Full Delay Shield + bonus	Rs 50 Cash + free delivery on next order, valid 48h

Abuse prevention

- Maximum 2 compensation events per user account per rolling 7-day window
- Minimum order subtotal of Rs 100 for Zomato Cash to be redeemable
- Delay Shield cash cannot be stacked with other active promo codes on the same order
- Accounts above 5 compensation events in 30 days are automatically flagged for review
- Credit is revoked if the order is cancelled within 3 minutes of the modal appearing

4. Success Metrics

North star

> Post-delay order rating >= 3.8/5.0 (current baseline: 2.1/5.0) Measured as the average in-app rating submitted by users who experienced an ETA breach > 5 minutes and received the Delay Shield notification. Aggregated weekly. Target: 3.8 or above within 60 days of full rollout.

Primary KPIs

Metric	Baseline	Target	Owner
Post-delay order rating	2.1 / 5.0	>=3.8 / 5.0	Consumer PM
Delay-related support tickets (% of total)	~34%	<22%	Support Ops
30-day retention after a delay event	~71%	>=82%	Growth PM
Delay-related share of negative reviews	48% of negatives	<28% of negatives	Data Team
Push notification open rate	N/A — new feature	>=68%	Growth PM
Modal CTA tap rate	N/A — new feature	>=80%	Consumer PM
Zomato Cash redemption within 24h	N/A — new feature	>=55% of credits issued	Payments PM

Guardrails

If any of the following move in the wrong direction at any point during rollout, the feature is paused immediately — regardless of primary KPI performance.

- Total compensation cost: must not exceed 1.2% of order GMV in the affected cohort
- Abuse rate: must stay below 0.4% of all compensated orders
- Push opt-out rate: must not rise by more than 2 percentage points post-launch
- Support ticket volume: must not increase — target is flat or declining
- Platform average delivery time: must remain flat — no operational degradation

5. A/B Test Plan

Hypothesis

Users who receive a proactive delay notification with automatic cash compensation will submit a meaningfully higher post-delivery rating than users who receive nothing. We expect an improvement of approximately +1.7 stars. Secondarily, treatment group users will be significantly less likely to raise a support ticket during or after a delay.

Test setup

Parameter	Detail
Groups	50% control (no action) vs 50% treatment (full Delay Shield)
Assignment unit	Per order — not per user — to prevent carry-over effects between sessions
Eligibility	ETA breach > 5 mins, subtotal >= Rs 100, status IN_TRANSIT, not already compensated
Duration	Minimum 3 weeks — eliminates day-of-week and promotional period effects
Minimum sample	1,000+ qualifying delayed orders per group (alpha=0.05, power=0.80)
Statistical test	Two-sample t-test on post-delay rating (primary). Chi-square on ticket rate (secondary).
Significance threshold	$p < 0.05$. Minimum detectable effect: +0.8 stars improvement in post-delay rating.

Decision rules

Outcome	Call
p<0.05, delta >=+0.8, all guardrails green	**Ship to 100%**
p<0.05, delta >=+0.8, but a guardrail breached	**Iterate — fix guardrail, retest at 5%**
p<0.05, delta <+0.8 — real but small effect	**Stakeholder call — ship for ticket savings?**
p>=0.05 — no significant effect	**Do not ship — post-mortem and redesign**
Any guardrail breached mid-test	**Pause immediately — investigate before resuming**

6. Edge Cases

The following cases were identified by walking through the trigger logic step by step. Each has a defined handling rule. Engineering should not need to make a judgment call on any of these.

Edge Case	Risk	Handling
Push notifications disabled	High	When user opens app during an active delay, show a floating in-app banner with the full modal content. Push is not required for this fallback path.
User already contacted support	High	Check for any open ticket in last 20 minutes before trigger fires. If found, suppress Delay Shield entirely. Let the support agent handle it.
Order cancelled after credit issued	Medium	Revoke the credit if order cancelled within 3 minutes of modal display. After 3 minutes, honour it — the user already experienced the delay.

User has two active orders	Medium	Include the restaurant name in both push copy and modal title. Example: "Your Bawa Chicken order is running late."
Reason API unavailable	Medium	Use hardcoded fallback: "There's an unexpected delay — we're on it." Never show blank, null, or a system error code to the user.
Weather surge — platform-wide delays	High	Operations toggle: during a SEVERE_WEATHER alert, compensation pauses automatically. Notification-only mode activates. Ops re-enables manually when conditions clear.
Low-value order abuse	Medium	Minimum Rs 100 subtotal for cash to apply. Cap of 2 events per user per 7 days. Auto-flag accounts above 5 events per month.
Poor network connection	Low	Modal uses pre-cached assets. Falls back to a simple static text banner if the rich component cannot load within 2 seconds.
Non-English app language	Medium	Localize push copy and modal in Hindi and the 6 regional languages with highest user representation. Use the language set in the user's app preferences.

7. Launch Plan

Phases

Phase	Timeline	Traffic	Gate to exit
0 — Engineering build	Weeks 1–3	Internal builds only	All tests passing, code reviewed, QA sign-off in writing
1 — Internal dogfood	Weeks 4–5	Zomato employees only	Zero P0/P1 bugs, cash credit success >99%, push delivery >95%
2 — Controlled beta	Weeks 6–7	5% — Bangalore and Mumbai only	All guardrails stable, cost within model, push open rate >=50%

3 — A/B test	Weeks 8–10	50/50 split across all Tier 1 cities	Statistical analysis complete, decision made per Section 5
4 — General availability	Week 11 onward	100% of all users nationally	Monitoring live, runbook published, 30-day review scheduled

What gets built in Phase 0

- Backend: ETA breach detection service — 60-second polling interval per active IN_TRANSIT order
- Backend: Delay Reason API integration with full fallback handling per Section 3.4
- Backend: Compensation ledger service with Zomato Cash wallet API integration and idempotency keys
- Backend: Abuse prevention rule engine — enforced at service layer, not application layer
- Frontend (Android + iOS): Delay Shield modal built pixel-perfect to the Figma spec in Section 3.2
- Frontend: In-app floating banner fallback for users with push disabled
- Analytics: modal_shown, cta_tapped, cash_credited, push_opened, ticket_suppressed, abuse_flagged
- QA: Full regression suite plus dedicated coverage for all 9 edge cases in Section 6

Risks

Risk	Likelihood	Contingency
Compensation cost exceeds budget	Medium	Raise ETA breach threshold from 5 to 15 minutes. Rerun cost model. Get Finance re-approval before resuming.
Delay Reason API unreliable	Medium	Default to UNKNOWN fallback. File reliability improvement ticket with Partner Engineering to run in parallel.
A/B test inconclusive — low delay volume	Low	Extend test to 5 weeks. Add selected Tier 2 cities to increase qualifying order sample.
Wallet API integration delayed	Low	Ship Tier 0 notification-only as fast path. Add cash compensation in v1.1 once wallet integration passes QA.

Stakeholders

Stakeholder	Role	Their concern
VP Consumer Product	Executive sponsor — approves each phase gate	User satisfaction metrics and compensation cost within budget
Engineering Lead	Technical delivery owner	Feasibility, timeline, edge case coverage
Data Science	A/B test design and result analysis	Metric integrity and honest interpretation of results
Support Operations	Primary beneficiary of ticket reduction	Ticket volume impact and agent training on the feature
Finance	Budget approver for compensation cost	GMV percentage staying within the approved threshold

Open Questions

- Should compensation be Zomato Cash or free delivery? User preference data would inform this decision.
- Does the 5-minute breach threshold feel too sensitive to the ops team? Needs calibration against real delay distribution data.
- Should restaurant-caused delays trigger the same compensation as partner-caused delays? Currently treated identically.
- Do we surface Delay Shield in onboarding or app store copy to set user expectations ahead of time?